

Enhanced Equivalent Circuit Model-Aided Broadband Single-Lossy-Single-Lossless-Layer Rasorber Design With High Selectivity

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Abstract—This letter presents a novel broadband single-lossy-single-lossless-layer frequency selective rasorber (FSR) with high passband selectivity facilitated by an enhanced equivalent circuit model (ECM). Leveraging the ECM method for an FSR by substituting typical LC resonant circuits with mixed series-parallel LCL resonant circuits for both the lossy and lossless FSSs significantly enhances the passband selectivity. The intentionally introduced inductance in the lossy FSS sharpens the passband roll-off, while in the lossless FSS creates an additional transmission zero, further improving selectivity. Simulations reveal a -3 dB passband of 10.04 GHz to 10.56 GHz, with a minimum insertion loss of 0.48 dB at 10.28 GHz. The proposed FSR achieves a 139.4% fractional bandwidth, spanning from 3.61 GHz to 20.22 GHz, and maintains stable performance for both TE and TM polarizations across a wide incidence up to 55° . The agreement between simulation and measurement confirms the practical application potential of the FSR design.

Index Terms—Frequency selective rasorber (FSR), high selectivity, mixed series-parallel LCL resonant circuit, wideband.

I. INTRODUCTION

DIFFERENT from conventional frequency selective surfaces (FSSs) [1], [2] and frequency selective absorbers [3], frequency selective rasorbers (FSRs) [4] provide a bandpass response within a specific frequency range while absorbing electromagnetic waves outside this range. This dual functionality effectively reduces target radar cross sections (RCS) and minimizes electromagnetic interference (EMI) from adjacent bands. Recent advancements have explored various types of FSRs, including transmission in front of an absorption band, transmission between two absorption bands, and behind an absorption band [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15]. Innovations in this field have achieved significant milestones such as broad absorption band [16], [17], [18], high angular stability [19], [20],

switchable passband [21], [22], dual passbands [23], [24], wide transmission band [25], [26], energy-selective passband [27], miniaturization [28], [29], and reconfigurability [30], [31].

Low selectivity, characterized by a gradual passband transition, results in relatively wide transition bandwidths with poor absorptive and non-transmissive properties. This impairs RCS reduction and degrades both absorption and transmission bandwidths [32]. To improve selectivity, several methods such as subwavelength resonance [33], second-order resonance [34], aperture coupling resonance [35], and multiorder passbands [36] have been investigated. These designs usually involve multiple lossless FSS layers, leading to an increase in design complexity and fabrication costs. Additionally, Minkowski fractal-shaped resonators [15] and 3-D structures [37] have been introduced as alternatives, whereas both require more complex configuration designs. Despite these achievements, the pursuit of improved selectivity continues to be a critical area of research to optimize the performance of frequency selective rasorbers.

In this letter, a new broadband highly selective FSR composed of a single-lossy-layer FSS and a single-lossless-layer FSS is introduced. An enhanced equivalent circuit model (ECM) is initially developed to facilitate precise design of the FSR for the desired selectivity. This is implemented by substituting the conventional LC resonant circuit with a mixed series-parallel LCL circuit. The inductance particularly introduced in the lossy FSS effectively sharpens the passband roll-off, while in the lossless FSS creates an extra transmission zero, which further enhances the selectivity. Simulation and experimental results demonstrate the high selectivity with angular stability up to 55° for both TE and TM polarizations.

II. ANALYSIS AND DESIGN OF THE RASORBER

A. Equivalent Circuit Design and Analysis

In general, an absorption-transmission-absorption (A-T-A) FSR exhibits a characteristic of both absorption and transmission patterns across its operational frequency band. This band is typically divided into a lower absorption band, an upper absorption band, and a transmission band in between. Fig. 1 shows the ECM of the FSR developed for achieving the proposed frequency responses. In addition to the necessary lossy and lossless layers, a dielectric matching layer (DML) is incorporated to enhance the bandwidth and incident angular stability. The transmission lines with various lengths and characteristic impedances represent different dielectric layers in the FSR. The characteristic

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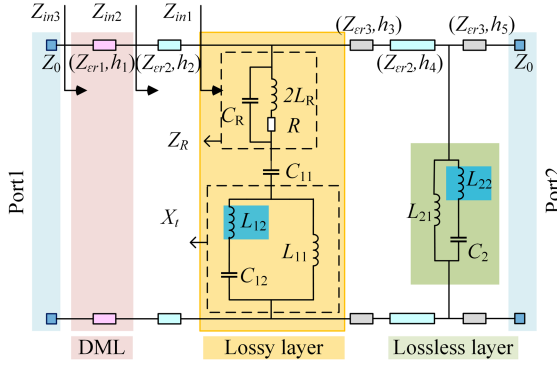


Fig. 1. Equivalent circuit model of the proposed FSR. ($R = 150 \, \Omega$, $L_R = 0.453 \, \text{nH}$, $C_R = 0.037 \, \text{pF}$, $C_{11} = 0.169 \, \text{pF}$, $C_{12} = 0.250 \, \text{pF}$, $C_2 = 0.283 \, \text{pF}$, $L_{11} = 0.46 \, \text{nH}$, $L_{12} = 0.5 \, \text{nH}$, $L_{21} = 0.48 \, \text{nH}$ and $L_{22} = 0.364 \, \text{nH}$. $Z_0 = 377 \, \Omega$, $Z_{er1} = 201 \, \Omega$, $Z_{er2} = 197 \, \Omega$, and $Z_{er3} = 368 \, \Omega$. h_x is the length of the transmission lines.)

impedances are calculated by $Z = Z_0 / \sqrt{\epsilon_{ri}}$, where $Z_0 = 377 \, \Omega$, $\epsilon_{r1} = 3.50$, $\epsilon_{r2} = 1.05$ and $\epsilon_{r3} = 3.66$.

The lossy layer in the FSR is responsible for transmitting electromagnetic power within the passband and dissipating it outside the passband. Consequently, the lossy layer is designed as an $Z_R C_{11} X_t$ series circuit (Fig. 1), where Z_R represents the equivalent circuit of the lumped resistors in a 0402 package. The ohmic loss induced by the $Z_R C_{11}$ series circuit facilitates the absorption performance outside the passband, while X_t is designed to obtain an infinite reactance at the transmission pole necessary for a passband. Typically, X_t composed of a parallel LC circuit, is used in an FSR design. However, such a design usually features a gradual roll-off of transmission. To address this, we introduced a mixed series-parallel $L_{12} C_{12} L_{11}$ circuit. By deliberately adding an extra L_{12} in series with C_{12} , higher selectivity and greater design flexibility in X_t can be achieved. Thus, by letting $X_t = \infty$, the transmission pole frequency (f_{tp}) can be predicted as follows:

$$f_{tp} = \frac{1}{2\pi\sqrt{(L_{11} + L_{12})C_{12}}}. \quad (1)$$

Keeping f_{tp} unchanged, the influences of inductance L_{12} and capacitance C_{12} on X_t and transmission coefficient (S_{21}) are analyzed, as depicted in Fig. 2. It is observed that increasing L_{12} and C_{12} quickly hone X_t [Fig. 2(a) and (b)]. Compared to the LC circuit in typical FSRs, the mixed series-parallel $L_{12} C_{12} L_{11}$ beneficially sharpens the roll-off of the passband, especially with larger values of L_{12} and C_{12} , as shown in Fig. 2(c). The reduction of transmission on both sides of the passband will lead to higher selectivity. Notably, the values of L_{12} and C_{12} are limited by the configuration of the lossy FSS. Therefore, $L_{12} = 0.5 \, \text{nH}$ and $C_{12} = 0.25 \, \text{pF}$ are finally determined.

To further enhance passband selectivity, another mixed series-parallel LCL circuit, representing the proposed single lossless FSS layer, is introduced with different inductance and capacitance values (Fig. 1). Notably, the lossless layer also functions as a reflecting ground for electromagnetic power absorption outside the passband. This mixed series-parallel $L_{22} C_2 L_{21}$ circuit creates both a transmission pole and a transmission zero. The transmission pole frequency of the lossless layer, determined by $1/(2\pi\sqrt{(L_{21} + L_{22})C_2})$, should be designed to match f_{tp} in (1) for ensuring the transmission of the FSR. Besides, its

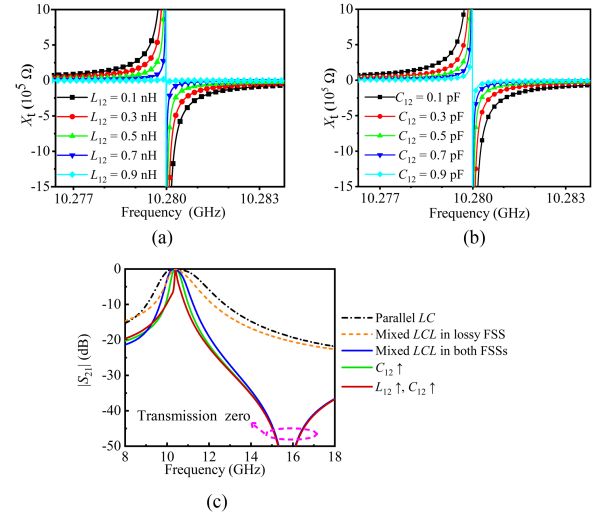


Fig. 2. Effects of (a) L_{12} and (b) C_{12} on the reactance X_t . (c) Comparison of $|S_{21}|$ for five cases. 1) Conventional parallel LC: $L = 0.96 \, \text{nH}$, $C = 0.25 \, \text{pF}$ for lossy layer and $L = 0.84 \, \text{nH}$, $C = 0.28 \, \text{pF}$ for lossless layer. 2) Proposed mixed LCL: $L_{12} = 0.5 \, \text{nH}$, $C_{12} = 0.2 \, \text{pF}$ for only one lossy FSS; 3) Proposed mixed LCL for both lossy and lossless FSSs. 4) Increased $C_{12} = 0.25 \, \text{pF}$ and 5) increased $L_{12} = 0.75 \, \text{nH}$ and $C_{12} = 0.25 \, \text{pF}$. Note that for the last three cases, the values of L_{22} , C_2 , and L_{21} for the lossless layer retain the same as in Fig. 1.

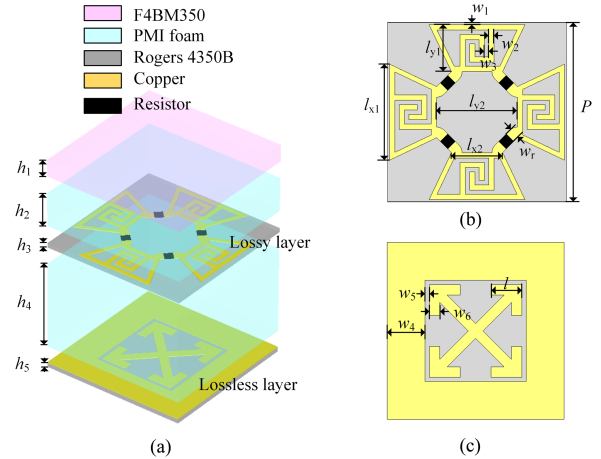


Fig. 3. Geometry of the proposed FSR. (Unit: mm, $P = 8.4$, $l = 1.47$, $l_{x1} = 4.5$, $l_{x2} = 2.4$, $l_{y1} = 2.2$, $l_{y2} = 3.8$, $w_1 = 0.1$, $w_2 = 0.25$, $w_3 = 0.2$, $w_4 = 1.8$, $w_5 = 0.2$, $w_6 = 0.5$, $w_7 = 0.5$, $h_1 = 1$, $h_2 = 2$, $h_3 = 0.254$, $h_4 = 5$, and $h_5 = 0.254$. $R = 147 \, \Omega$.)

transmission zero frequency f_{tz} can be derived from

$$f_{tz} = \frac{1}{2\pi\sqrt{L_{22}C_2}}. \quad (2)$$

This transmission zero greatly suppresses the high-frequency transmission beyond the passband, thereby further enhancing the passband selectivity, as shown in Fig. 2(c).

B. FSR Design and Analysis

Fig. 3 illustrates the unit cell geometry configuration of the designed FSR, crafted to achieve the desired performance metrics discussed above. The FSR consists of three layers: a lossy FSS, a lossless FSS, and a dielectric matching layer (DML),

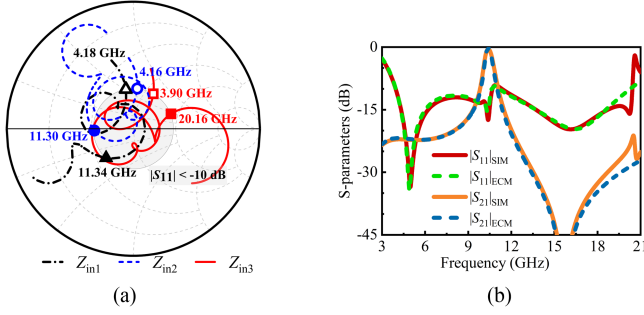


Fig. 4. (a) Impedance matching with DML and (b) S -parameters of the FSR.

separated by polymethacrylimide (PMI) foams with a dielectric constant of 1.05.

The lossy FSS comprises four symmetric trapezoidal loops with interdigital-coupled lines. These loops are interconnected with strip lines embedded with four lump resistors, as shown in Fig. 3(b). Unlike common rectangular loops, the trapezoidal pattern increases the capacitance C_{11} formed by the gaps between adjacent unit cells, ensuring effective low-frequency absorption. The specially designed interdigital-coupled lines are capable of generating a large inductance L_{12} and a large capacitance C_{12} in series, necessary for achieving high passband selectivity as aforementioned. The lumped resistor R , connected in series with C_{11} and X_t , is used to produce two absorption bands on both sides of the passband.

To generate the mixed series-parallel $L_{22}C_2L_{21}$ circuit, the lossless FSS is designed with a cross-arrow pattern for L_{22} and a grid for L_{21} , while the gaps between them create the capacitance C_2 , as shown in Fig. 3(c). It is worth noting that the lossless layer in this design utilizes a single-layer structure, rather than a combination of multiple FSS layers. This lossless FSS contributes to the same bandpass response as the lossy layer while introducing an additional transmission zero, resulting in sharper passband roll-off. Both the lossy and lossless layers are printed on the Rogers 4350B substrate, which has a relative permittivity of 3.66 and a loss tangent of 0.0037.

Additionally, a DML made of F4BM350 with a relative permittivity of 3.5 and a loss tangent of 0.001 is placed above the lossy FSS, separated by a PMI foam. The normalized impedances of the FSR with and without DML are depicted in the impedance Smith chart in Fig. 4(a), covering a frequency range from 3.5 GHz to 20.5 GHz. The comparison reveals observable improvement in impedance matching for the FSR with DML, especially in the frequency range above 11.34 GHz.

Directly placing an F4BM350 slab on top of the lossy layer will fail to obtain impedance matching in a wide frequency band, since the impedance curve shown in Fig. 4(a) rotates around the center of $0.53Z_0(Z_0/\sqrt{3.5})$. To resolve this, a PMI foam is introduced in between them, transforming the impedance from Z_{in1} to Z_{in2} . However, the PMI foam primarily affects the phase of the reflection coefficient, while having minimal impact on its amplitude, resulting in only a little increase of -10 dB bandwidth. The dielectric layer of F4BM350 is then added on the PMI foam, further transforming the impedance Z_{in2} to Z_{in3} , as depicted in Fig. 4(a). It is obvious that most of the curve of Z_{in3} shifts into the -10 dB region, indicating better impedance matching over a wide frequency range.

The S -parameters of the FSR derived from ECM have been compared with those simulated using CST [38] under normal

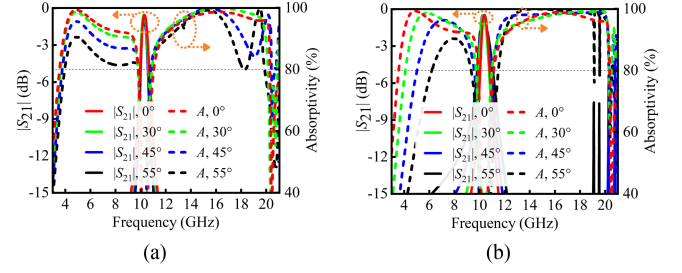


Fig. 5. Comparison of the FSR performance between measurement and simulation for (a) TE and (b) TM polarization.

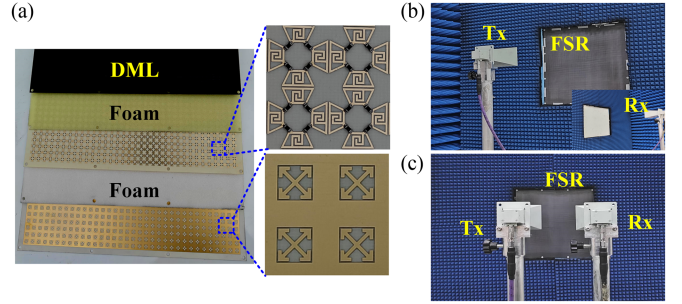


Fig. 6. (a) Photograph of fabricated FSR and experimental setup for (b) transmission and (c) reflection measurements.

incidence, as shown in Fig. 4(b). Note that the package structures of the lumped resistors are incorporated in the simulation model. Good agreements observed validate the ECM analysis.

C. Simulated Performance of the Proposed Rasorber

The simulated $|S_{21}|$ and absorptivity of the proposed FSR under normal and oblique incidence up to 55° for both TE and TM polarizations are illustrated in Fig. 5, incorporating higher-order Floquet modes in the simulation. Under normal incidence, the passband ($|S_{21}| > -3$ dB) ranges from 10.04 GHz to 10.56 GHz with a minimum insertion loss (IL) of 0.48 dB at 10.28 GHz. The quality factor (Q) is 19.80, representing the selectivity of the FSR, which is quantitatively defined as the ratio of center frequency to -3 dB transmission bandwidth. The lower and upper absorption bands (Absorptivity $> 80\%$) cover the frequency range of 3.61 GHz to 9.82 GHz and 11.03 GHz to 20.22 GHz, respectively. The transition bands ($|S_{21}| < -3$ dB, Absorptivity $< 80\%$) between the passband and the absorption band have fractional bandwidths of 2.2% [(9.82 to 10.04) GHz] and 4.4% [(10.56 to 11.03) GHz]. These narrow transition bands indicate pretty good selectivity of the FSR.

As the incident angle increases up to 55° , the frequency response for TE-polarization remains stable, regardless of a slight increase of the minimum IL to 0.97 dB. A discontinuity occurs at a high frequency of around 18 GHz, attributed to the harmonics of the interdigital-coupled resonator in the lossy layer. However, this discontinuity has a negligible impact on the overall performance of the designed FSR. For TM-polarization, the minimum IL remains 0.48 dB, while the absorption band shifts towards higher frequency, attributed to the impedance mismatch incurred by the increase of incident angle. Notably, the first grating lobe appears around 19.6 GHz, contracting the high absorption band to 11.5 GHz to 19.1 GHz for both polarizations.

TABLE I
PERFORMANCE COMPARISON TO RELATED STUDIES

Ref.	PB ¹ / Insertion Loss	LAB ² (GHz)	UAB ³ (GHz)	Unit Size (mm)	Thickness (mm)	Q^4	AS ⁵
[9]	9.5 GHz / 0.26 dB	3.45 - 8.7 (86.4%)	10.33 - 16.1 (43.7%)	11 (0.13 λ_L)	10 (0.12 λ_L)	13.57	45°
[15]	8.4 GHz / 0.7 dB	3.4 - 7.5 (75.2%)	9.2 - 10.8 (16%)	15 (0.17 λ_L)	9.6 (0.10 λ_L)	16.25	40°
[29]	10 GHz / 0.25 dB	3.84 - 9.21 (82.3 %)	10.91 - 13.9 (24.1%)	12 (0.15 λ_L)	6 (0.08 λ_L)	16.95	40°
[34]	4.3 GHz / 0.08 dB	1.83 - 4.06 (75.7%)	4.56 - 6.44 (34.2%)	25 (0.15 λ_L)	16.5 (0.10 λ_L)	13.44	40°
[35]	4.65 GHz / 0.12 dB	2.1 - 4 (50%)	5.47 - 8.43 (42.3%)	9.6 (0.07 λ_L)	11.1 (0.08 λ_L)	11.63	45°
[36]	7.66 - 11.76 GHz / 1 dB	2.93 - 6.57 (76%)	12.52 - 15.43 (20.8%)	15 (0.15 λ_L)	12.5 (0.12 λ_L)	15.3%; 6.26% [#]	30°
[37]	3.02 GHz / 1 dB	/	5.37 - 8.59 (46.13%)	12(/)	36(/)	56% [#]	30°
This work	10.28 GHz/0.48 dB	3.61 - 9.82 (92.5%)	11.03 - 20.22 (58.8%)	8.5 (0.10 λ_L)	8.4 (0.10 λ_L)	19.80 3.46%; 5.80%[#]	55°

¹PB: Passband frequency, except for cut-off frequency in [37]. ²LAB, ³UAB: Lower and upper absorption bandwidth with absorptivity above 80%, except [34] and [37] above 90%. ⁴ Q : Center frequency / (-3 dB transmission bandwidth), except fractional transition bandwidth ($|S_{21}| < -1$ dB, Absorptivity < 80%) in [36] and [37]. ⁵AS: Angular stability.

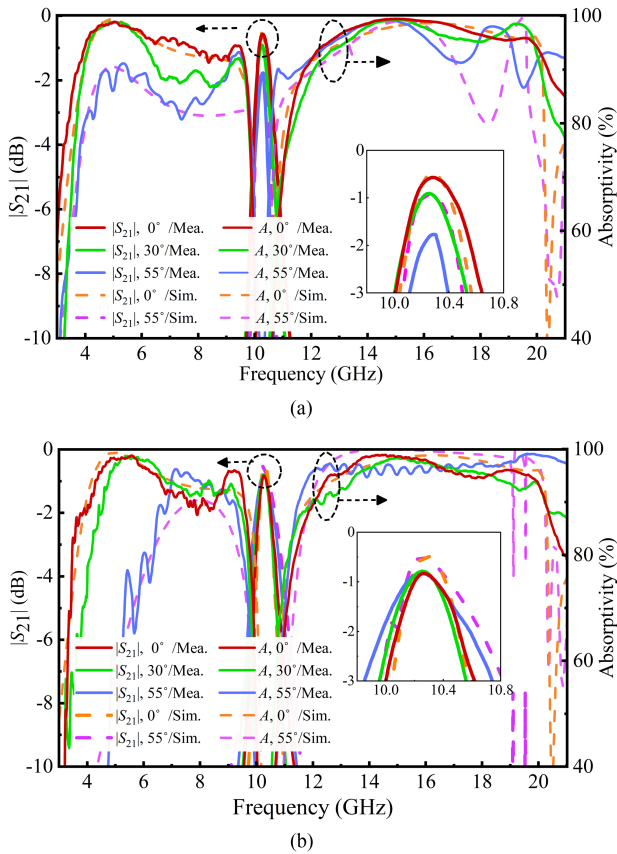


Fig. 7. Comparison of the FSR performance between measurement and simulation for (a) TE and (b) TM polarization.

III. EXPERIMENTAL MEASUREMENT

To further verify the absorptive and transmissive performance of the proposed FSR, a prototype consisting of 35×35 elements with a dimensional size of $300 \text{ mm} \times 300 \text{ mm}$ was fabricated, as shown in Fig. 6(a)–(c). A total of 4900 chip resistors [39] were soldered on the lossy layer. All layers of the FSR were assembled by nylon screws and nuts. A free space measurement method [3], [39] was employed to obtain the S -parameters. Two pairs of broadband horn antennas operating at 2 GHz to 18 GHz and 8 GHz to 26.5 GHz are connected to an Agilent E8363C

PNA network analyzer for measurement, respectively, as shown in Fig. 6.

The measurement results of the proposed FSR, depicted in Fig. 7, are compared to the simulation results. A strong correlation can be observed. For normal incidence, the operating frequency bands cover 3.46 GHz to 20.95 GHz and 3.45 GHz to 21.43 GHz for TE and TM polarizations, respectively. A highly selective passband ranging from 10.00 GHz to 10.55 GHz is obtained, with a minimum IL of 0.57 dB (TE) and 0.83 dB (TM) at 10.28 GHz. The Q factors calculated from the measured results are 18.71 (TE) and 20.12 (TM). When the incident angle reaches 55°, the measured minimum IL increases to about 1.7 dB (TE) and 1.1 dB (TM). The main reasons for discrepancies between the measurement and simulation stem from fabrication tolerance, variations in lumped element values with frequency, and the limited size of the measured sample.

The performance of the proposed FSR is also compared to some related works, as shown in Table I. It is evident that the proposed absorber exhibits a higher Q factor and narrower fractional transition bandwidth, indicating higher selectivity. Besides, the FSR offers better performance, including wider absorption bandwidths and better angular stability.

IV. CONCLUSION

In this letter, a highly selective FSR based on single-lossy-single-lossless-layer FSSs has been introduced. By replacing the conventional parallel LC circuits with mixed series-parallel LCL circuits in ECM to characterize the desired responses, the passband selectivity can be significantly enhanced using only a single lossy FSS layer and a single lossless FSS layer. The mechanism behind this improvement is analyzed, followed by a meticulous design of the FSR configuration to fulfill the required inductances and capacitances in both lossy and lossless FSSs. Finally, a DML is incorporated to improve the bandwidth and incident angular stability. The FSR manifests a transmission window centered at 10.28 GHz with a minimum IL of 0.48 dB and an impressive Q factor of up to 19.80. The 80% absorption bands adjacent to the passband range from 3.61 GHz to 9.82 GHz and from 11.03 GHz to 20.22 GHz, with an angular stability of 55° for both TE and TM polarizations. Measurement results are in good accordance with the simulations, affirming the effectiveness of the proposed design.

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